
Mitigating Bias in Low-SNR Financial Reinforcement Learning via Quantum Representations

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Abstract

The financial market is a typical low Signal-to-Noise Ratio (SNR) setting, which can usually destabilize off-policy maximum-entropy methods like Soft Actor-Critic (SAC). Specifically, noisy state representations may produce unreliable Q-value estimates, and bootstrapping amplifies these errors, i.e., a failure mode we call the “Financial Entropy Trap”. In this paper, we propose FPQC-SAC, an efficient and plug-and-play SAC variant that places a compact and bounded Parameterized Quantum Circuit (PQC) before the actor and critic networks to constrain feature propagation at the representation level, rather than filtering raw inputs or regularizing Q-values after bootstrapping. Notably, FPQC-SAC reduces the impact of extreme market fluctuations on Bellman target estimation, while trainable quantum entanglement preserves flexible cross-asset interactions. Empirical evaluations on real-world portfolio management tasks demonstrate that FPQC-SAC substantially enhances out-of-sample stability and cumulative returns by achieving a 66.89% relative gain in cumulative return over standard unconstrained SAC and outperforms the best continuous-control deep reinforcement learning baseline by $\sim 27\%$.

1 Introduction

Financial markets are inherently low signal-to-noise ratio (SNR) environments. Useful financial signals are often extremely weak, whereas short-term fluctuations, market microstructure noise, investor sentiment, and liquidity shocks are dominant. Prior studies have shown that financial prediction naturally suffers from these high noises, along with non-stationarity and distribution shifts. This makes models prone to fitting historical noise patterns, consequently suffering severe performance degradation in out-of-sample testing or real trading environments [16, 8, 7, 45]. Under such extreme uncertainty, sequential financial decision-making faces profound challenges.

At the core of this challenge lies sequential asset allocation: continuously determining when and how much to buy, hold, or sell. While traditional statistical models struggle to characterize complex market-state transitions, Deep Reinforcement Learning (DRL) provides a promising end-to-end framework [40, 39, 22, 46, 29]. Among these, **maximum-entropy reinforcement learning**, represented by Soft Actor-Critic (SAC) [38], explicitly maximizes policy entropy to encourage stochastic exploration, aiming to maintain trading-policy diversity and generate high-quality decisions.

However, the Actor-Critic architecture becomes significantly fragile in noise-dominant markets. Entropy-driven stochastic exploration does not necessarily gather effective information; instead, it may amplify noise-dominated update directions. This prevents the policy from stably capturing weak predictive signals. Consequently, noise contaminates value estimation and guides erroneous exploration, forming a self-reinforcing loop. We refer to this critical failure mode of maximum-entropy RL in financial decision-making as the *Financial Entropy Trap*.

36 To escape the Financial Entropy Trap, we first examine existing stabilization mechanisms. We
 37 consider two direct solutions: transferring general RL stabilization methods, and introducing explicit
 38 input filtering. Unfortunately, both paradigms exhibit inherent limitations:

- 39 • **Output-side regularization intervenes too late.** Methods like clipped double-Q learning
 40 [15], large-scale critic ensembles [41], and distributional critic truncation [5] mainly act
 41 at the end of the update pipeline. Even contemporary advancements utilizing dropout Q-
 42 functions [19] or cross-Q updates [9] focus primarily on stabilizing network-layer gradients
 43 during Bellman bootstrapping. Once market noise has passed through conventional encoders
 44 and contaminated intermediate representations [12, 37], posterior corrections can only
 45 provide *after-the-fact* mitigation. It becomes nearly impossible to recover weak predictive
 46 signals that have already been mixed or distorted.
- 47 • **Input-side filtering intervenes too rigidly.** Classical input-denoising pipelines, such as
 48 Kalman filtering, wavelet denoising, and Fourier-domain filtering, attempt to directly sup-
 49 press noise before states enter the model [1, 4, 35]. However, these methods usually rely on
 50 strong handcrafted priors, e.g., treating noise as measurement uncertainty or approximating
 51 it as high-frequency perturbations. Because useful financial signals themselves may also
 52 be transient and high-frequency, aggressive filtering often erases scarce valuable signals
 53 alongside the noise.

54 Neither approach fundamentally blocks the propagation of noise perturbations into value estimation
 55 while preserving weak financial signals.

56 To bridge this gap, we draw inspiration from a structurally relevant physical phenomenon in quantum
 57 optics [21], where destructive interference partially cancels noise fluctuations without degrading
 58 the primary signal. This provides a crucial insight for low-SNR RL: effective noise suppression
 59 should not force a rigid choice between isolated pre-filtering or post-clipping, but rather introduce
 60 an incremental, structural constraint during signal representation. Inspired by this, we propose the
 61 **FPQC-SAC** architecture by embedding a Parameterized Quantum Circuit (PQC) at the representation
 62 front-end of SAC. As a plug-and-play enhancement, PQC goes beyond simply adding a new module; it
 63 leverages the natural interference structure induced by entangled evolution and measurement readout.
 64 This allows unstructured noise to statistically cancel within a high-dimensional representation space,
 65 effectively decoupling noise suppression from signal retention and overcoming the limitations of
 66 classical approaches.

67 The main contributions of this paper are summarized as follows:

- 68 • **Identification of the Financial Entropy Trap and the Representation Gap.** We theoret-
 69 ically characterize a critical failure mode of maximum-entropy RL in low-SNR environments,
 70 demonstrating how stochastic exploration systematically amplifies market noise. Further-
 71 more, we provide a principled analysis of why existing filtering and regularization methods
 72 fail to balance noise suppression with information preservation.
- 73 • **Proposal and Theoretical Grounding of the FPQC-SAC Architecture.** Moving beyond
 74 heuristic module combinations, we design FPQC-SAC by embedding a Parameterized Quan-
 75 tum Circuit at the SAC front-end. We theoretically motivate how quantum entanglement and
 76 interference inherently provide a structural constraint that dynamically cancels unstructured
 77 noise while preserving temporally correlated weak signals, offering a rigorous decoupling
 78 mechanism that traditional architectures lack.
- 79 • **Empirical validation and mechanism explanation.** Rigorous backtesting on three real-
 80 world datasets shows that FPQC-SAC improves performance over standard SAC by at
 81 least 50% and outperforms the best baseline model by more than 20%. Ablation studies
 82 further verify that this performance leap originates from the trainable quantum entanglement
 83 structure and trainable quantum rotation matrices.

84 2 Related Work

85 2.1 Financial RL for Portfolio Management and Continuous Control

86 In quantitative finance, continuous portfolio management and asset allocation are canonically for-
 87 mulated as Markov Decision Processes (MDPs). The development of standardized open-source

environments, such as FinRL [40] and FinRL-Meta [39], has established deep reinforcement learning (DRL) as a prevalent paradigm for data-driven trading. Researchers have extensively evaluated both on-policy methods, such as Proximal Policy Optimization (PPO), and off-policy continuous control algorithms, including DDPG, TD3, and SAC, across various market configurations [22, 46]. While on-policy methods are often favored by practitioners for their stability and robustness in highly volatile markets, they inherently sacrifice the sample efficiency and deep exploration capabilities provided by off-policy architectures [29]. Restoring the training stability of off-policy continuous control methods in financial environments remains a critical challenge.

2.2 Overestimation Bias and Instability in Noisy Off-Policy Learning

A primary vulnerability of off-policy Q-learning is the overestimation bias, which arises from the maximization operation over noisy Q-value estimates during Bellman bootstrapping [15]. In continuous control, TD3 addresses this via clipped double-Q learning to establish a pessimistic lower bound [15], while SAC introduces maximum entropy regularization to encourage robust exploration [38]. Subsequent advancements, such as REDQ [41] and TQC [5], utilize massive network ensembles or distributional critics to actively truncate excessive overestimation.

However, when applied to financial time series characterized by inherently low signal-to-noise ratios (SNR), these value-level heuristic clipping mechanisms often prove insufficient. The root cause extends beyond algorithmic bootstrapping: high-capacity neural encoders exhibit representation vulnerability. Operating strictly within unbounded Euclidean spaces, these classical networks inadvertently map infinite-variation Brownian spatial perturbations directly into the latent space [12, 37]. Consequently, unbounded spatial noise bypasses the shared encoders and compromises the downstream critic ensembles. This observation motivates our focus on structural interventions at the geometric representation bottleneck level.

2.3 Quantum RL for Continuous Control

The advent of data re-uploading techniques has enabled Parameterized Quantum Circuits (PQCs) to function as universal approximators for continuous variables, bypassing the limitations of the no-cloning theorem [2]. This capability has catalyzed the extension of Quantum Reinforcement Learning (QRL) into complex continuous control tasks. Early architectures and empirical applications have demonstrated competitive performance across various physical domains [3, 32, 17]. Furthermore, quantum policy gradient methods have been theoretically and empirically shown to accelerate learning convergence [34, 43].

Nevertheless, the predominant focus of these existing QRL explorations has been uniformly constrained to parameter efficiency and sample complexity within relatively deterministic physical simulators. Even in recent attempts to port hybrid QRL into financial trading, the quantum layer is typically relegated to serving as a generic nonlinear feature map [44]. In this work, we pivot the application of QRL: rather than primarily optimizing for parameter reduction, we propose utilizing the quantum measurement process as a structural representation bias tailored to limit the propagation of noise under extreme low-SNR Itô financial environments.

2.4 PQC Expressivity, Trainability, and Geometric Bottlenecks

The design of quantum neural networks is strictly governed by the trade-off between expressibility and trainability. Theoretical studies highlight that excessive circuit depth leads to highly expressive representations that rapidly induce barren plateaus [26, 13]. In the context of noisy financial data, excessive expressibility introduces an additional hazard: potential overfitting to stochastic variance. Our architecture actively circumvents this by adopting a shallow PQC design. By treating the restricted capacity of shallow circuits as a feature rather than a limitation, we enforce an intentional geometric bottleneck. This unitary-constrained configuration naturally restricts the model’s ability to fit arbitrary Brownian diffusion [31], inherently regularizing the representation and fostering robust generalization in out-of-sample portfolio evaluations.

136 3 Methodology and Theoretical Foundations

137 This section systematically establishes why the quantum representation layer provides a robust struc-
 138 tural mechanism to mitigate Q-value overestimation within the Soft Actor-Critic (SAC) framework.
 139 We first formalize the extreme low Signal-to-Noise Ratio (SNR) dynamics of financial markets as
 140 stochastic differential equations (SDEs) governed by high-dimensional Brownian motion, clarifying
 141 the core representational difficulty. Crucially, we shift the paradigm from classical value-clipping
 142 to geometric interference. We introduce the network topology and progressively analyze the core
 143 structural effects of the Parameterized Quantum Circuit (PQC) bottleneck—hyperspherical phase
 144 encoding, entanglement-driven isotropic noise cancellation, and constructive signal alignment—
 145 demonstrating how this architecture intrinsically dissipates Temporal Difference (TD) target variance
 146 rather than merely enforcing hard boundaries.

147 3.1 Low-SNR Itô SDE and the Representation Challenge

148 In the continuous-time limit, the state variables or asset prices in financial markets can be modeled as
 149 multi-dimensional Itô diffusion processes.

150 **Assumption 1** (Low-SNR Local Itô Transition). *Let the state vector $X_t \in \mathbb{R}^d$ evolve according to*
 151 *the following stochastic differential equation (SDE):*

$$dX_t = \mu(X_t, A_t)dt + \sigma(X_t)dW_t \quad (1)$$

152 *where $\mu(\cdot)$ is the drift function carrying the directional predictive signal, $\sigma(\cdot)$ is the instantaneous*
 153 *volatility matrix, and W_t is a d -dimensional standard Brownian motion. In the financial low-SNR*
 154 *regime, the critical characteristic is that the magnitude of the drift is vastly eclipsed by the diffusion*
 155 *term:*

$$\|\mu(X_t, A_t)dt\| \ll \|\sigma(X_t)dW_t\| \quad (2)$$

156 Although the Brownian increment dW_t possesses independent increments and a zero mean, its
 157 sample paths exhibit infinite variation over any arbitrarily small time interval. This implies that on
 158 the discrete timescales of algorithmic observation and policy updates, the spatial perturbation of state
 159 X_t is mathematically unbounded in Euclidean space.

160 The direct consequence of this formulation is catastrophic for standard deep reinforcement learning.
 161 If unconstrained raw states are fed directly into classical Critic networks, the high-variance diffusion
 162 term is injected unmitigated into value function estimation. This causes the bootstrapped Bellman
 163 Temporal Difference (TD) targets to fluctuate violently. This mechanism is the representational
 164 root of the *financial entropy trap*: noise inherently pollutes the state representation, distorts the
 165 Q-value gradients, and is progressively amplified by self-bootstrapping. We provide a mathematical
 166 formalization of the **Financial Entropy Trap** in **Appendix 6.3.2**.

167 3.2 FPQC-SAC Architecture and End-to-End Workflow

168 To fundamentally resolve the Financial Entropy Trap, we must separate the weak drift μ from the
 169 dominating diffusion σdW_t . Simply clipping the input or bounding the output is insufficient, as it
 170 rigidly squashes both noise and signal. Instead, we embed a Parameterized Quantum Circuit (PQC)
 171 as a shared representation bottleneck to naturally filter noise via complex hyperspherical interference.
 172 The end-to-end data flow is detailed in Figure 1.

173 Hybrid Quantum-Classical Architecture and Hyperspherical Isolation

174 The framework maps high-noise financial observations (A_t, X_t) onto a stabilized latent manifold.
 175 The state space is initially compressed via a classical linear layer to a 7-dimensional vector, aligning
 176 with the qubit topology. These features are subsequently embedded into the quantum Hilbert space
 177 via continuous R_x angle encoding. The unitary evolution $U(X, \theta)$ utilizes $L = 2$ layers of circular
 178 CNOT entanglement and single-qubit rotations. The resulting latent representation f_t is derived from
 179 the expectation values of Pauli-Z observables, as formalized in Definition 1.

180 **Definition 1** (PQC Latent Mapping). *Let $U(X, \theta) = \prod_{l=1}^L W_l(\theta_l)R_x(X)$ denote the parameterized*
 181 *quantum circuit. The latent representation $f(X, \theta) \in \mathbb{R}^k$ is extracted via the expected values of Pauli*
 182 *observables $\{\hat{O}_j\}_{j=1}^k$ following Born's rule:*

$$f_j(X, \theta) = \text{Tr} \left(\hat{O}_j U(X, \theta) |0\rangle \langle 0| U^\dagger(X, \theta) \right). \quad (3)$$

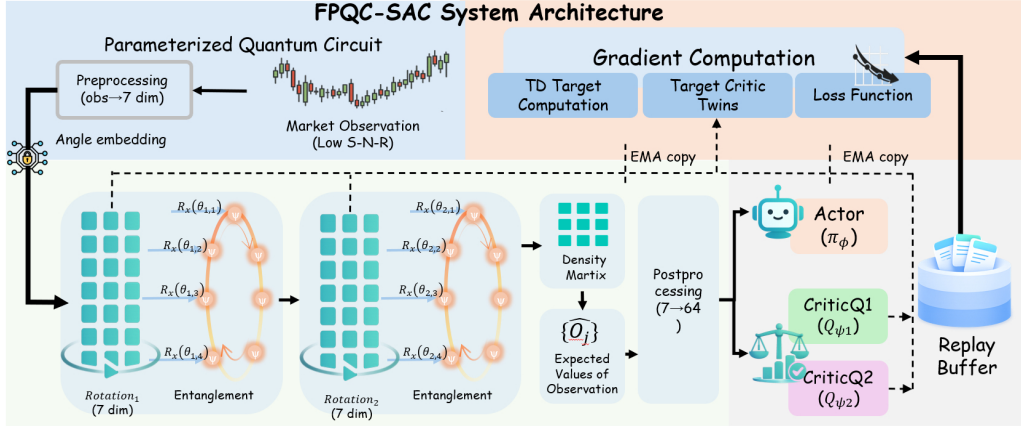


Figure 1: The system architecture and end-to-end workflow of FPQC-SAC. By forcing high-variance irregular Brownian paths through a 7-qubit, 2-layer circularly entangled unitary space, the architecture maps Euclidean noise onto a complex hypersphere. Here, isotropic noise undergoes destructive interference, naturally filtering extreme perturbations before they reach the classical Actor-Critic heads.

183 In the reinforcement learning phase, the latent features f_t serve as inputs for the stochastic Actor
 184 policy π_ϕ and the Twin Critics $Q_{\psi_{1,2}}$. Crucially, an Exponential Moving Average (EMA) copy of the
 185 PQC parameters ($\bar{\theta}$) is included within the target network cascade. This ensures that the Bellman
 186 bootstrapping process evaluates states strictly through the lens of the quantum interference bottleneck,
 187 preventing raw environmental variance from infiltrating the non-smooth $\min(Q_{\bar{\psi}_1}, Q_{\bar{\psi}_2})$ operator.
 188 Detailed explanations of this interference-driven noise suppression are provided in Sections 3.3, 3.4,
 189 and 3.5.

190 3.3 Expectation Attenuation via Complex Phase Encoding

191 Before evaluating the quantum encoding, we must account for the classical Pre-Net. As formally
 192 proven in **Appendix ?? (Lemma 6.3.3)**, passing the raw Itô diffusion through a piecewise-linear,
 193 locally Lipschitz Pre-Net (e.g., ReLU) functionally preserves the local Itô structure almost everywhere,
 194 mapping the effective noise covariance to $\Sigma_Z = J_h \Sigma_X J_h^\top$. The unbounded nature of the Brownian
 195 noise propagates into the latent space, setting the stage for PQC attenuation.

196 The first layer of the PQC’s denoising mechanism originates from its fundamental operation of
 197 angle encoding (the R_x gates shown in Stage 3), which maps classical real-valued inputs into
 198 complex exponential phases. Let the pre-processed latent state evolve according to the Itô process
 199 $dZ_s = \mu_s ds + \sigma_s dW_s$ over the time interval $s \in [t, t + \Delta t]$.

200 **Proposition 1** (Diffusion Attenuation in Periodic Complex Features). *Given an initial latent state Z_t ,
 201 and assuming the drift μ_s and diffusion covariance $\Sigma_s = \sigma_s \sigma_s^\top$ are locally stable over $[t, t + \Delta t]$,
 202 encoding the Itô diffusion into a continuous unitary generator mode ω yields the following expected
 203 evolution of the complex phase:*

$$\mathbb{E}[e^{i\omega^\top Z_{t+\Delta t}}] = e^{i\omega^\top Z_t} \cdot \underbrace{\exp\left(i \int_t^{t+\Delta t} \omega^\top \mu_s ds\right)}_{\text{Signal Phase Translation}} \cdot \underbrace{\exp\left(-\frac{1}{2} \int_t^{t+\Delta t} \omega^\top \Sigma_s \omega ds\right)}_{\text{Diffusion Attenuation}}. \quad (4)$$

204 This analytical equation (detailed proof provided in **Appendix 6.3.1**) mathematically decouples two
 205 distinctly divergent effects:

- 206 • **Signal Preservation:** The predictive drift term μ_s is perfectly preserved as a directional
 207 phase translation. The directional information of the causal signal survives cleanly within
 208 the argument of the complex exponential.

209 • **Diffusion Attenuation:** Due to its independent increments and zero mean, the stochastic
 210 diffusion $\sigma_s dW_s$ algebraically induces an exponential decay factor weighted by the covari-
 211 ance Σ_s under expectation. Intuitively, the isotropic random jumps of Brownian motion
 212 are “rotationally averaged” across the complex plane—random phase oscillations in all
 213 directions mutually cancel out in expectation.

214 The core mathematical insight here is that the complex exponential mapping naturally separates the
 215 deterministic trend of the signal from the random diffusion of the noise in an expected sense. Signals
 216 are retained as phase shifts, while noise is mathematically suppressed as an attenuation factor.

217 3.4 The Core Role of Entanglement: Synergistic Signal-to-Noise Separation

218 The true filtering capability of the PQC does not stem merely from its finite output range, but from
 219 how entanglement structurally discriminates between noise and signal within the hypersphere. This
 220 functionality manifests in two complementary dimensions:

221 **Scattering Noise into Isotropic Dimensions (Destructive Interference).** If the PQC solely relied on
 222 complex phase encoding without inter-qubit interactions, noise would remain isolated within single-
 223 qubit subspaces. Introducing the circular CNOT chain instantaneously diffuses local, independent
 224 Brownian perturbations into a collective, multi-body entangled state. Because the financial diffusion
 225 term σdW_t is stochastic and largely unstructured, entanglement scrambles these errors into a high-
 226 dimensional, isotropic phase perturbation. Due to the projective nature of Pauli measurements (the
 227 trace operation), these uniformly scattered random phases destructively interfere and statistically
 228 cancel each other out upon readout. Classical sine/cosine bases cannot replicate this synergistic,
 229 high-dimensional cancellation.

230 **Aligning Signals along Principal Directions (Constructive Interference).** While noise is scattered
 231 isotropically, the weak drift signal μdt possesses intrinsic temporal and cross-asset correlations (e.g.,
 232 subtle momentum continuations). Under the tuning of trainable rotation gates, the circuit does not
 233 scatter these correlated phases uniformly. Instead, it dynamically rotates the hyperspherical basis to
 234 align these coherent phase evolutions along specific Pauli measurement directions. Ablation results in
 235 Table 4 demonstrate that removing CNOT gates or freezing rotations destroys performance. Simply
 236 put, entanglement treats noise and signal asymmetrically: it scrambles isotropic noise into destructive
 237 cancellation, while combing the structural signal into a focused, constructively interfering output.
 238 For a detailed mathematical analysis of these entangled modes, see Appendix 6.3.4.

239 3.5 Geometric Consequence: Variance Attenuation via Hyperspherical Cancellation

240 To establish a mathematically rigorous safeguard for off-policy learning, we formalize the conse-
 241 quence of the aforementioned phase interference. While classical bottlenecks (like Tanh) merely clip
 242 extreme values—leaving the variance dense at the boundaries—the PQC fundamentally attenuates
 243 the variance of the underlying stochastic process.

244 **Proposition 2** (Hyperspherical Phase Cancellation and Variance Attenuation). *Under unitary evo-*
 245 *lution, the state ρ is confined to a norm-preserving complex hypersphere. When subjected to the*
 246 *unbounded local Itô diffusion dW_t , the spatial variance is mapped to phase fluctuations. During*
 247 *the expected measurement over Pauli observables, isotropic phase variations lacking structural*
 248 *correlation undergo destructive interference. Consequently, for an input spatial perturbation with*
 249 *variance Σ_X , the variance of the resulting PQC representation, $\text{Var}(f_{\text{PQC}})$, is exponentially attenu-*
 250 *ated relative to the intensity of the isotropic noise, completely decoupling the output stability from the*
 251 *infinite-variation nature of the raw SDE.*

252 Because this attenuation derives geometrically from the trace operation over an entangled complex
 253 manifold, it provides a superior structural guarantee compared to classical bounds.

254 **Corollary 1** (Stabilized Representation Cascade). *In the hybrid FPQC-SAC architecture, even if*
 255 *the classical Pre-Net amplifies the raw market noise, the subsequent projection through the PQC*
 256 *bottleneck forces the isotropic components to destructively cancel. Assuming a locally Lipschitz Post-*
 257 *Net, the final representation perturbation $\|f(X) - f(\bar{X})\|_2$ exhibits a tightly concentrated variance,*
 258 *immune to the catastrophic Brownian spikes that bypass standard Euclidean neural encoders.*

259 In this context, the quantum bottleneck acts not as a simple hard-clipping wall, but as an active
 260 representational dissipater that absorbs and neutralizes high-variance unstructured noise.

3.6 Structural Regularization of TD Targets via Variance Dissipation

We now connect this geometric cancellation to the most noise-sensitive component of SAC: the Temporal Difference (TD) target. The vulnerability of standard maximum-entropy RL lies in the operator $y_t = r_t + \gamma \max_{a'} Q(X_{t+1}, a')$. When X_{t+1} contains unmitigated Brownian noise, the maximization over noisy representations systematically induces a positive bias, inflating the Q-values.

In our framework, the bootstrapped target is computed as:

$$y_t = r_t + \gamma \mathcal{B}(f(X_{t+1})), \quad (5)$$

where $\mathcal{B}$ denotes the locally Lipschitz soft Bellman target map.

Because $f(X_{t+1})$ has passed through the PQC, its isotropic noise components have been dissipated via destructive interference (Proposition 2). Consequently, the variance of the input to the Bellman operator is drastically suppressed: $\text{Var}(f(X_{t+1})) \ll \text{Var}(X_{t+1})$.

This means the amplitude of the spurious spikes entering $\mathcal{B}(\cdot)$ is structurally minimized. Without extreme positive noise spikes, the max (or soft-max) operator has no artificial high values to exploit. Thus, FPQC-SAC systematically starves the overestimation bias of its necessary fuel (high-variance perturbations). By dissipating the noise *before* Bellman bootstrapping rather than trying to regularize the Q-values *after* the fact, the architecture successfully bypasses the amplification channel of the Financial Entropy Trap.

3.7 Closing the Loop: Theoretical Mechanisms Meet Empirical Validation

The mechanisms elucidated in this section—complex hyperspherical encoding, entanglement-driven destructive/constructive interference, and variance dissipation—are structurally unified. Phase encoding converts spatial noise to phase variation, entanglement scrambles it isotropically, and quantum measurement neutralizes it while preserving the coherently aligned drift signals.

To rigorously guarantee that this interference-based filtering does not squash causal signals alongside the noise, **Appendix 6.4** derives a theoretical lower bound for the Output Signal-to-Noise Ratio (SNR_{out}). We mathematically demonstrate that while unstructured noise variance is aggressively dissipated, the sensitivity to deterministic structural signals remains strictly bounded away from zero.

This mathematically justified mechanism formulates a verifiable hypothesis: the PQC acts as an interference-driven filter that separates signal from noise, thus stabilizing value estimation. Therefore, Section 4 evaluates FPQC-SAC through a complete empirical causal chain:

$$\begin{aligned} &\text{Interference-driven signal-preserving bottleneck} \\ &\rightarrow \text{Suppressed TD variance and minimal Q-gap} \\ &\rightarrow \text{Superior out-of-sample trading performance.} \end{aligned} \quad (6)$$

Specifically, we first test whether the PQC bottleneck can suppress TD-target variance and reduce the Q-gap without destroying structural information (outperforming simple classical bounded activation like Tanh). Subsequently, we evaluate whether this intrinsic Bellman stability directly translates into superior out-of-sample trading returns. Through this closed loop of theoretical mechanisms, algorithmic stability, and final financial performance, the superiority of the hyperspherical cancellation paradigm is thoroughly validated.

4 Experiments

4.1 Experimental Setup

We evaluate FPQC-SAC on three contrasting U.S. multi-asset portfolios: (1) an extremely low-SNR portfolio (AAPL, AMZN, GOOGL, MSFT, QQQ, SPY) with high volatility and stochastic interference; (2) a stable blue-chip portfolio (BAC, JNJ, JPM, PG, WMT, XOM) with persistent signals and low noise; and (3) a high-growth technology portfolio (AVGO, BRK.B, META, NFLX, NVDA, TSLA) with concentrated growth exposure and elevated regime sensitivity. All DRL agents are trained from 2013-01-02 to 2018-12-31 for 25,000 environment steps and tested out-of-sample from 2019-01-02 to 2021-08-31, covering the COVID-19 shock and recovery. Metrics are reported as $\text{mean} \pm \text{std}$ over independent seeds using a 10% two-sided trimmed mean, removing the two

highest and two lowest cumulative-return seeds from 20 runs. Detailed parameters are provided in Appendix 6.2. Baselines are divided into three categories: (1) classic continuous-control DRL algorithms (SAC, TD3, PPO, DDPG, A2C); (2) output-regularized robust DRL methods for mitigating Q-value overestimation (REDQ, TQC); (3) rule-based trading strategies, including MACD, SMA, KDJ-RSI, and ZMR.

4.2 Q-Value Overestimation Mitigation and Bottleneck Analysis

In this section, we respond to the theoretical hypothesis proposed in Section 3.7 by testing whether the PQC representation constraint can suppress TD-target variance and reduce the Q-gap. During the out-of-sample testing phase, we measured the deviation between the learned Critic values and the Monte Carlo soft returns (Q-gap), alongside Temporal Difference variance (TD-var) and Critic-loss variance (C-loss-var). Furthermore, to definitively rule out the alternative hypothesis that "any structural representation bottleneck could achieve similar stabilizing effects," we systematically substituted the PQC with a suite of classical representation designs. These include bounded activations (Tanh), feature clipping (Clipped Latent), dimensionality compression (Linear and MLP Bottlenecks), generic regularization (Weight Decay), and normalization techniques (LayerNorm and SpectralNorm). Each bottleneck is embedded within the identical Actor-Critic backbone and trained under the exact same protocol. Table 1 reveals key findings. Unconstrained SAC suffers catastrophic Q-value explosions

Table 1: Systematic comparison between classic representation bottlenecks and PQC (FPQC-SAC) shows that FPQC-SAC achieves the best return-stability metrics.

Model	CR% $\uparrow$	AR% $\uparrow$	SR $\uparrow$	Sortino $\uparrow$	Calmar $\uparrow$	Q-gap $\downarrow$	TD-var $\downarrow$	C-loss-var $\downarrow$
Linear Bottleneck [30]	98.9 $\pm$ 60.1	27.8 $\pm$ 16.8	1.18 $\pm$ 0.63	1.31 $\pm$ 0.75	1.06 $\pm$ 0.56	-0.553 $\pm$ 2.16	4.16 $\times$ 10 ³ $\pm$ 7.49 $\times$ 10 ³	4.71 $\times$ 10 ⁵ $\pm$ 1.20 $\times$ 10 ⁶
SAC (No Bottleneck) [38]	117.0 $\pm$ 18.6	33.7 $\pm$ 4.29	1.40 $\pm$ 0.12	1.59 $\pm$ 0.19	1.11 $\pm$ 0.23	26.4 $\pm$ 81.8	2.55 $\times$ 10 ⁷ $\pm$ 1.01 $\times$ 10 ⁸	1.32 $\times$ 10 ¹³ $\pm$ 4.78 $\times$ 10 ¹³
Weight Decay Bot. [23]	125.0 $\pm$ 33.4	35.4 $\pm$ 7.54	1.43 $\pm$ 0.19	1.71 $\pm$ 0.26	1.31 $\pm$ 0.32	-0.148 $\pm$ 0.30	23.6 $\pm$ 31.6	157 $\pm$ 277
MLP Bottleneck [20, 18]	136.0 $\pm$ 29.2	37.9 $\pm$ 6.38	1.53 $\pm$ 0.18	1.77 $\pm$ 0.30	1.31 $\pm$ 0.33	-0.047 $\pm$ 0.14	25.1 $\pm$ 35.1	140 $\pm$ 334
SpectralNorm Bot. [27]	144.0 $\pm$ 65.2	38.4 $\pm$ 16.4	1.44 $\pm$ 0.57	1.73 $\pm$ 0.71	1.38 $\pm$ 0.60	1.92 $\times$ 10 ¹² $\pm$ 8.73 $\times$ 10 ¹²	2.91 $\times$ 10 ²⁷ $\pm$ 1.30 $\times$ 10 ²⁸	2.99 $\times$ 10 ⁵⁷ $\pm$ 1.34 $\times$ 10 ⁵⁸
Tanh Bottleneck [24]	154.0 $\pm$ 30.4	41.6 $\pm$ 6.35	1.57 $\pm$ 0.16	1.88 $\pm$ 0.20	1.46 $\pm$ 0.25	-0.004 $\pm$ 0.05	0.81 $\pm$ 1.02	14.8 $\pm$ 28.4
Clipped Latent [11, 42]	155.0 $\pm$ 30.1	41.9 $\pm$ 6.20	1.55 $\pm$ 0.21	1.91 $\pm$ 0.19	1.54 $\pm$ 0.20	-0.007 $\pm$ 0.05	0.93 $\pm$ 1.65	18.7 $\pm$ 46.6
LayerNorm Bot. [6]	170.0 $\pm$ 36.8	44.9 $\pm$ 7.52	1.63 $\pm$ 0.21	2.00 $\pm$ 0.26	1.64 $\pm$ 0.29	-0.002 $\pm$ 0.05	0.76 $\pm$ 0.74	8.99 $\pm$ 14.3
FPQC-SAC (Ours)	195.3$\pm$26.9	50.0$\pm$5.15	1.71$\pm$0.15	2.13$\pm$0.19	1.75$\pm$0.20	0.0004$\pm$0.05	0.54$\pm$0.69	8.19$\pm$13.7

and extreme estimation variance, which distort policy gradients and mislead the actor. To address this, our analysis focuses fundamentally on the role of bounded features. Bounded constraints themselves are not rare. Classical operations, e.g., tanh activations, gradient clipping, and LayerNorm can also impose range restrictions on feature spaces. While these classical bottlenecks exhibit a clear return-stability trade-off—with LayerNorm and Tanh variants achieving moderate returns and partial stability gains—they cannot fully suppress noise while retaining weak predictive signals. Furthermore, we note that some ill-designed representation constraints exhibit extremely high estimation variance, which is primarily attributed to catastrophic training seed explosions.

Only PQC simultaneously delivers high returns and stability. Crucially, addressing our theoretical propositions from the Methodology, FPQC-SAC achieves the absolutely lowest TD-variance (0.54), Q-gap (0.0004), and Critic-loss-variance (8.19) across all tested architectures. It significantly outperforms the most competitive classical bounded baselines, further reducing the TD-variance compared to LayerNorm (0.76) and Tanh (0.81), while achieving a much tighter Q-gap approximation than Clipped Latent (0.007). These stability advantages seamlessly translate into performance, allowing FPQC-SAC to reach a 195.3% cumulative return, comfortably outperforming the runner-up LayerNorm Bottleneck (170.0%). Ultimately, PQC’s unique combination of unitary-constrained bounded readout and entanglement-induced phase cancellation creates a stable architecture unreplicable by conventional designs. For more detailed variance, explained-variance, Q-gap, and TD-var analyses of representative spectral/filtering encoders, please see Appendix 6.1.

4.3 Trading Performance Comparison

Having established strong empirical evidence that the PQC bottleneck fundamentally stabilizes value estimation (via reduced TD-variance and Q-gap in Section 3.2), we now complete the causal chain: evaluating whether this algorithmic stability seamlessly translates to out-of-sample trading efficacy. To ensure rigorous evaluation and fair comparison, we structure our backtesting into one primary experiment based on standard domain benchmarks, followed by two supplementary generalization experiments.

For the **Main Experiment**, we evaluate the models on the canonical 6-asset portfolio (AAPL, AMZN, GOOGL, MSFT, QQQ, SPY) widely established by the FinRL[40] benchmark. To ensure the validity of the experiment, we have additionally extended the length of the test set in the original FinRL[40] benchmark. This explicitly aligns our evaluation with the classical baseline standards in the financial RL community, representing a robust allocation environment that reflects the core beta of the U.S. market and ensuring an apples-to-apples comparison with existing literature. While general continuous-control baselines (e.g., PPO, A2C, DDPG) are included for comprehensiveness, our primary focus is a systematic benchmark against the maximum-entropy SAC family. By comparing FPQC-SAC with vanilla SAC, early output-regularized variants (REDQ, TQC), and contemporary regularized architectures (DroQ [19], CrossQ [9]), we ensure a thorough chronological evaluation to demonstrate the superiority of representation-level constraints over downstream heuristic regularizations.

While many financial RL studies limit their evaluation to a single standard benchmark, evaluating off-policy algorithms in highly stochastic environments inherently demands broader context. Therefore, to rigorously assess the out-of-sample generalization capability across divergent market regimes, we additionally introduce two contrasting supplementary portfolios: **Sub-experiment 1 (Defensive Blue-chip)** consists of traditional economy and value stocks (BAC, JNJ, JPM, PG, WMT, XOM), presenting a low-noise, highly persistent signal environment. **Sub-experiment 2 (High-Volatility Growth)** consists of aggressive thematic assets (NVDA, META, AVGO, TSLA, NFLX, BRK.B), providing an extremely volatile environment designed to stress-test models against severe stochastic interference. For these sub-experiments, we strictly focus on off-policy continuous control baselines. All metrics follow the standard 10% Trimmed Mean protocol over 20 random seeds.

Table 2: Main Experiment: Comprehensive performance comparison on the Mainstream Tech & Market-Index Portfolio (QQQ, SPY, MSFT, GOOGL, AMZN, AAPL). Red marks the best performance and blue marks the second-best.

Category	Model	CR% $\uparrow$	AR% $\uparrow$	SR $\uparrow$	Sortino $\uparrow$	Calmar $\uparrow$
Market	B&H	88.57 $\pm$ 0.00	26.90 $\pm$ 0.00	1.18 $\pm$ 0.00	1.28 $\pm$ 0.00	0.80 $\pm$ 0.00
Our Model	FPQC-SAC	195.32$\pm$26.94	50.05$\pm$5.15	1.71$\pm$0.15	2.13$\pm$0.19	1.75$\pm$0.20
DRL-based	SAC [38]	117.04 $\pm$ 18.56	33.67 $\pm$ 4.29	1.40 $\pm$ 0.12	1.59 $\pm$ 0.19	1.11 $\pm$ 0.23
	CrossQ [9]	148.19 $\pm$ 38.63	40.32 $\pm$ 8.21	1.54 $\pm$ 0.26	1.87 $\pm$ 0.34	1.53 $\pm$ 0.36
	DroQ [19]	152.80 $\pm$ 21.61	41.55 $\pm$ 4.51	1.56 $\pm$ 0.13	1.91 $\pm$ 0.15	1.56 $\pm$ 0.18
	REDQ [41]	147.87 $\pm$ 23.48	40.48 $\pm$ 5.16	1.57 $\pm$ 0.14	1.85 $\pm$ 0.20	1.44 $\pm$ 0.24
	TQC [5]	145.10 $\pm$ 23.24	39.89 $\pm$ 5.01	1.57 $\pm$ 0.11	1.82 $\pm$ 0.18	1.38 $\pm$ 0.24
	TD3 [15]	151.55 $\pm$ 29.20	41.20 $\pm$ 6.12	1.57 $\pm$ 0.12	1.87 $\pm$ 0.21	1.46 $\pm$ 0.29
	DDPG [25]	153.65$\pm$27.25	41.67$\pm$5.68	1.60$\pm$0.12	1.90$\pm$0.20	1.48$\pm$0.28
	PPO [33]	151.94 $\pm$ 27.50	41.31 $\pm$ 5.72	1.56 $\pm$ 0.15	1.89 $\pm$ 0.20	1.48 $\pm$ 0.21
Rule-based	A2C [28]	145.76 $\pm$ 21.32	40.06 $\pm$ 4.57	1.51 $\pm$ 0.14	1.84 $\pm$ 0.16	1.46 $\pm$ 0.20
	MACD	14.55 $\pm$ 0.00	5.23 $\pm$ 0.00	0.39 $\pm$ 0.00	0.38 $\pm$ 0.00	0.28 $\pm$ 0.00
	SMA	49.47 $\pm$ 0.00	16.29 $\pm$ 0.00	0.87 $\pm$ 0.00	0.83 $\pm$ 0.00	0.70 $\pm$ 0.00
	KDJ_RSI	0.86 $\pm$ 0.00	0.32 $\pm$ 0.00	0.74 $\pm$ 0.00	0.17 $\pm$ 0.00	0.58 $\pm$ 0.00
	ZMR	43.12 $\pm$ 0.00	14.41 $\pm$ 0.00	0.82 $\pm$ 0.00	0.85 $\pm$ 0.00	0.52 $\pm$ 0.00

Table 3: Generalization Experiments: Assessing the stability and adaptability of off-policy baselines across two contrasting regimes. Red marks the best performance and blue the second-best.

Category	Model	Defensive Blue-chip Portfolio (Sub-experiment 1)					High-Volatility Growth Portfolio (Sub-experiment 2)				
		CR% $\uparrow$	AR% $\uparrow$	SR $\uparrow$	Sortino $\uparrow$	Calmar $\uparrow$	CR% $\uparrow$	AR% $\uparrow$	SR $\uparrow$	Sortino $\uparrow$	Calmar $\uparrow$
Our Model	FPQC-SAC	45.01$\pm$13.58	14.86$\pm$4.20	0.62$\pm$0.21	0.81$\pm$0.28	0.64$\pm$0.33	183.26$\pm$75.58	46.77$\pm$14.20	1.37$\pm$0.30	1.76$\pm$0.39	1.35$\pm$0.36
Off-policy DRL	SAC [38]	27.25 $\pm$ 24.62	8.97 $\pm$ 8.45	0.41 $\pm$ 0.35	0.51 $\pm$ 0.44	0.33 $\pm$ 0.29	133.33 $\pm$ 54.18	36.72 $\pm$ 11.37	1.14 $\pm$ 0.33	1.46 $\pm$ 0.39	1.10 $\pm$ 0.29
	CrossQ [9]	47.15$\pm$6.11	15.59$\pm$1.79	0.62$\pm$0.11	0.76$\pm$0.13	0.50$\pm$0.13	87.31 $\pm$ 28.66	26.26 $\pm$ 7.13	1.02 $\pm$ 0.22	1.20 $\pm$ 0.28	0.87 $\pm$ 0.26
	DroQ [19]	31.71 $\pm$ 7.42	10.86 $\pm$ 2.33	0.50 $\pm$ 0.09	0.60 $\pm$ 0.12	0.35 $\pm$ 0.07	111.37 $\pm$ 40.43	31.93 $\pm$ 9.43	1.11 $\pm$ 0.30	1.37 $\pm$ 0.34	1.01 $\pm$ 0.25
	REDQ [41]	41.47 $\pm$ 15.48	13.75 $\pm$ 4.96	0.58 $\pm$ 0.22	0.74 $\pm$ 0.28	0.49 $\pm$ 0.22	136.65 $\pm$ 82.74	38.20 $\pm$ 16.31	1.10 $\pm$ 0.30	1.50 $\pm$ 0.37	1.10 $\pm$ 0.27
	TQC [5]	30.02 $\pm$ 31.07	9.56 $\pm$ 10.74	0.44 $\pm$ 0.41	0.54 $\pm$ 0.54	0.39 $\pm$ 0.38	118.14 $\pm$ 69.60	32.58 $\pm$ 15.75	1.15 $\pm$ 0.43	1.42 $\pm$ 0.54	1.02 $\pm$ 0.41
	TD3 [15]	33.68 $\pm$ 11.91	11.42 $\pm$ 3.77	0.51 $\pm$ 0.18	0.62 $\pm$ 0.21	0.35 $\pm$ 0.14	154.99$\pm$58.16	41.36$\pm$11.78	1.36$\pm$0.28	1.70$\pm$0.35	1.25$\pm$0.26
	DDPG [25]	35.85 $\pm$ 11.70	12.10 $\pm$ 3.63	0.55 $\pm$ 0.19	0.67 $\pm$ 0.23	0.41 $\pm$ 0.20	144.49 $\pm$ 39.61	39.45 $\pm$ 9.00	1.30 $\pm$ 0.25	1.64 $\pm$ 0.30	1.21 $\pm$ 0.21
Rule-based	MACD	4.11 $\pm$ 0.00	1.52 $\pm$ 0.00	0.25 $\pm$ 0.00	0.29 $\pm$ 0.00	0.11 $\pm$ 0.00	22.16 $\pm$ 0.00	7.81 $\pm$ 0.00	0.83 $\pm$ 0.00	0.95 $\pm$ 0.00	0.71 $\pm$ 0.00
	SMA	5.16 $\pm$ 0.00	1.91 $\pm$ 0.00	0.30 $\pm$ 0.00	0.29 $\pm$ 0.00	0.14 $\pm$ 0.00	37.34 $\pm$ 0.00	12.65 $\pm$ 0.00	0.99 $\pm$ 0.00	0.97 $\pm$ 0.00	0.80 $\pm$ 0.00
	KDJ_RSI	-1.82 $\pm$ 0.00	-0.69 $\pm$ 0.00	-0.12 $\pm$ 0.00	-0.16 $\pm$ 0.00	-0.05 $\pm$ 0.00	1.09 $\pm$ 0.00	0.41 $\pm$ 0.00	0.10 $\pm$ 0.00	0.08 $\pm$ 0.00	0.06 $\pm$ 0.00
	ZMR	10.90 $\pm$ 0.00	3.96 $\pm$ 0.00	0.24 $\pm$ 0.00	0.28 $\pm$ 0.00	0.13 $\pm$ 0.00	66.79 $\pm$ 0.00	21.18 $\pm$ 0.00	1.10 $\pm$ 0.00	1.21 $\pm$ 0.00	0.73 $\pm$ 0.00

Tables 2 and 3 demonstrate that FPQC-SAC consistently dominates across varied market regimes, explicitly highlighting the structural vulnerabilities of standard SAC improvements. In the **Main Experiment**, contemporary baselines successfully improve sample efficiency and stability over vanilla SAC, with DroQ [19] (152.80% CR) strongly competing with prior algorithms. Nevertheless, FPQC-SAC maintains a substantial 42-percentage-point lead (**195.32% CR**). This confirms that weak-signal-preserving representation filtering is fundamentally more effective than downstream heuristic regularizations in standard market conditions.

This divergence becomes highly nuanced in the generalization tests. In the defensive **Sub-experiment 1**, where noise is low and trends are persistent, standard off-policy algorithms typically struggle. Here, CrossQ [9] adapts exceptionally well, leveraging its cross-Q updates to fit weak trends and achieve the highest absolute return (**47.15% CR**). However, this emphasizes a classic risk-reward trade-off: while CrossQ extracts slightly higher absolute profits, FPQC-SAC trades off a marginal amount of return to retain significantly superior downside protection, dominating the risk-adjusted metrics with the highest Sortino (**0.81**) and Calmar (**0.64**) ratios. This validates the PQC bottleneck’s inherent capability to restrict tail risks even in calm markets.

Most critically, **Sub-experiment 2** perfectly illustrates the *Financial Entropy Trap*. In this highly volatile portfolio, extreme stochastic noise severely penalizes traditional continuous control architectures. Despite their theoretical stability in standard benchmarks, both DroQ and CrossQ suffer severe structural collapse here, dropping to 111.37% and 87.31% CR respectively, completely underperforming vanilla SAC (133.33% CR). This exposes a fatal vulnerability: network-layer perturbations—such as dropout Q-functions [19] or cross-Q updates [9]—inadvertently amplify gradient variance when exposed to extreme out-of-distribution market noise. In stark contrast, FPQC-SAC effectively anchors policy updates via its path-wise bounded quantum readout, shrugging off the extreme variance to achieve a staggering **183.26% CR**. These results validate our core hypothesis: while downstream regularizations provide completeness and efficiency in standard settings, embedding a hard structural constraint at the representation level is the only mechanism that provides universal robustness across all financial regimes.

4.4 Ablations and Analysis

This part conducts ablation experiments to identify the core components driving performance gains. The fully trainable FPQC-SAC is benchmarked against two degraded variants: **Frozen-PQC FPQC-SAC**, which freezes all trainable rotation parameters while preserving the CNOT topology, and **No-CNOT FPQC-SAC**, which removes all CNOT connections whilst retaining trainable single-qubit rotations. We further vary PQC depth across $L \in \{1, 2, 3, 4, 5\}$ to explore the influence of circuit expressivity.

Table 4 shows disabling either parameter trainability or quantum entanglement markedly degrades performance: the cumulative return falls from 195.32% to 144.51% for Frozen-PQC and 148.49% for No-CNOT. The No-CNOT variant also presents substantially higher seed-level variance, indicating entanglement boosts both trading returns and training stability. These outcomes confirm the performance gain stems not from fixed bounded projection or isolated single-qubit rotations alone, but from the joint effect of trainable rotations and entangled circuit topology.

Table 4 further shows $L = 2$ yields the best overall performance: $L = 1$ underperforms, while $L \geq 3$ causes consistent degradation. This indicates FPQC-SAC benefits from a shallow yet expressive quantum bottleneck, with sufficient capacity to model useful market structure without overfitting stochastic fluctuations. Combined with representation analysis and classical bottleneck comparisons, these ablations confirm FPQC-SAC’s improvement stems from its specific shallow, trainable, entangled PQC design—not generic dimensionality reduction or bounded feature compression.

Table 4: Ablation studies of FPQC-SAC. Red/blue denote best/second best.

Model	CR% $\uparrow$	AR% $\uparrow$	SR $\uparrow$	Sortino $\uparrow$	Calmar $\uparrow$
Core Structural Ablation					
Trainable	195.32$\pm$26.94	50.05$\pm$5.15	1.71$\pm$0.15	2.13$\pm$0.19	1.75$\pm$0.20
No-CNOT	148.49 $\pm$ 69.60	40.37 $\pm$ 15.49	1.45 $\pm$ 0.38	1.86 $\pm$ 0.52	1.54 $\pm$ 0.43
Frozen-PQC	144.51 $\pm$ 20.21	39.76 $\pm$ 4.31	1.51$\pm$0.10	1.81 $\pm$ 0.16	1.39 $\pm$ 0.19
PQC Depth Ablation					
L1	150.90 $\pm$ 22.09	41.06 $\pm$ 4.75	1.48 $\pm$ 0.15	1.87 $\pm$ 0.18	1.52 $\pm$ 0.21
L2	195.32$\pm$26.94	50.05$\pm$5.15	1.71$\pm$0.15	2.13$\pm$0.19	1.75$\pm$0.20
L3	152.18 $\pm$ 21.83	41.33 $\pm$ 4.68	1.51 $\pm$ 0.12	1.90 $\pm$ 0.17	1.53 $\pm$ 0.20
L4	164.85 $\pm$ 37.17	43.73 $\pm$ 7.52	1.54 $\pm$ 0.19	1.94 $\pm$ 0.23	1.60 $\pm$ 0.29
L5	151.48 $\pm$ 27.71	41.15 $\pm$ 5.97	1.54$\pm$0.17	1.89 $\pm$ 0.22	1.48 $\pm$ 0.22

5 Conclusion

This paper addresses a theoretical vulnerability when applying maximum-entropy off-policy reinforcement learning to low-SNR financial environments and identifies the "Financial Entropy Trap," a theoretical vulnerability where maximum-entropy RL resonantly amplifies low-SNR noise into systematic Q-value overestimation, and establishes that a bounded quantum representation acts as a geometric firewall under local Lipschitz assumptions. Based on this, FPQC-SAC is proposed, an architectural innovation that rely on a shallow Parameterized Quantum Circuit (PQC) as a structural information bottleneck to map noisy inputs into periodic complex phases for statistical attenuation while leveraging circular entanglement to induce destructive interference of unstructured noise and constructively align weak predictive signals within a bounded unitary space. Extensive empirical evaluations validate this design: FPQC-SAC achieves a 66.89% relative gain in cumulative return over standard SAC, outperforms the strongest continuous-control DRL baseline by 27%, and yields the lowest TD-target variance (0.54). Ultimately, these results underscore a promising paradigm shift: utilizing quantum circuits not merely for computational speedup, but as robust geometric constraints to mitigate representation vulnerabilities in highly stochastic domains.

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6 Appendix

6.1 Representation Quality Evaluation

To verify whether PQC fulfills its core design goal of noise suppression and signal structure preservation, we analyze the variance and VAF of the learned representations. Under identical experimental settings, we extract representations from the same raw input state and report two groups of comparisons: representative spectral/filtering examples, including FPQC-SAC, RFF/Fourier features [36], Wavelet filtering [4], and Kalman filtering [1], and the classical bottleneck controls used in Table 1.

560 Total Latent Variance quantifies the overall intensity of spatial noise perturbations in the representa-
561 tion manifold, while VAF@1 (Variance Accounted For by the 1st PC) assesses structural coherence
562 and the representation’s capability to linearly preserve predictable environmental structure, with
563 higher values implying fewer unstructured noise components.

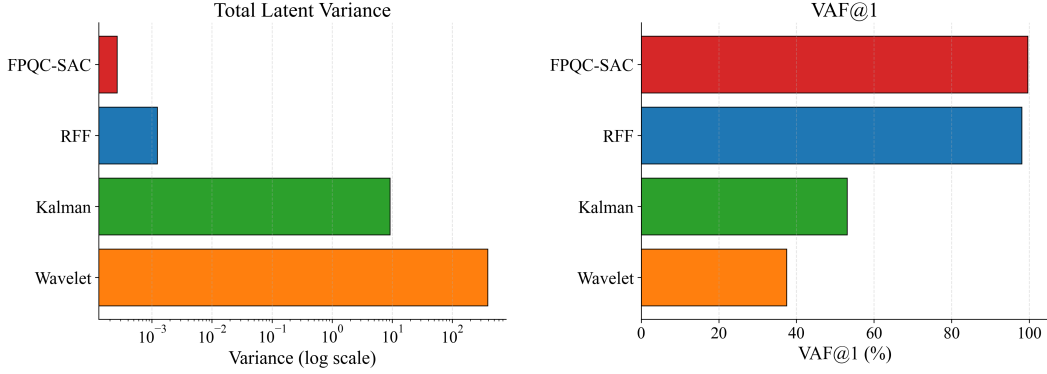


Figure 2: Representation quality evaluation for FPQC-SAC, RFF/Fourier features, Wavelet filtering, and Kalman filtering. The left panel reports Total Latent Variance on a log scale, where lower values indicate stronger spatial noise suppression. The right panel reports VAF@1, where higher values denote stronger predictive structural coherence.

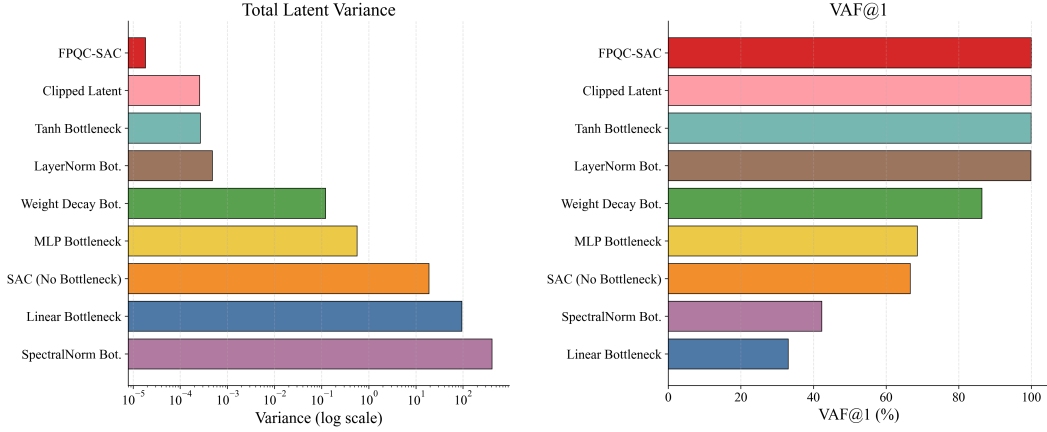


Figure 3: Representation quality evaluation for the classical bottleneck controls in Table 1. The left panel reports Total Latent Variance on a log scale, and the right panel reports VAF@1. Lower variance and higher VAF@1 indicate stronger noise suppression and more coherent latent structure.

Table 5: Performance and available Q-value training diagnostics for FPQC-SAC and representative signal-processing encoders.

Encoder	CR% $\uparrow$	AR% $\uparrow$	SR $\uparrow$	Sortino $\uparrow$	Calmar $\uparrow$	Q-gap $\downarrow$	TD-var $\downarrow$	C-loss-var $\downarrow$
FPQC-SAC	195.32 $\pm$ 26.94	50.05 $\pm$ 5.15	1.71 $\pm$ 0.15	2.13 $\pm$ 0.19	1.75 $\pm$ 0.20	$4.55 \times 10^{-4} \pm 4.61 \times 10^{-2}$	0.535$\pm$0.69	8.19$\pm$14
RFF	162.23 $\pm$ 37.25	43.30 $\pm$ 7.76	1.53 $\pm$ 0.22	1.94 $\pm$ 0.25	1.61 $\pm$ 0.29	$5.79 \times 10^{-4} \pm 5.10 \times 10^{-2}$	0.552 $\pm$ 0.67	8.08 $\pm$ 12
Wavelet	138.44 $\pm$ 36.39	38.23 $\pm$ 8.00	1.49 $\pm$ 0.17	1.76 $\pm$ 0.27	1.32 $\pm$ 0.35	-0.252 $\pm$ 0.37	$7.81 \times 10^2 \pm 1.74 \times 10^3$	$2.61 \times 10^3 \pm 4.96 \times 10^3$
Kalman	130.44 $\pm$ 23.78	36.66 $\pm$ 5.26	1.50 $\pm$ 0.15	1.72 $\pm$ 0.23	1.23 $\pm$ 0.26	-0.170 $\pm$ 0.05	10.00 $\pm$ 7.62	61.09 $\pm$ 95.59

564 6.2 Reproducibility Details

565 6.2.1 Markov Decision Process (MDP) and Environment Settings

566 The trading environment is formulated as a continuous-control Markov Decision Process (MDP). The
567 portfolio consists of 6 highly volatile U.S. assets: AAPL, AMZN, GOOGL, MSFT, QQQ, and SPY. In

Table 6: State space, action space, and environment configuration.

Parameter	Value / Description
State Space Dimension	64
State Composition	[1 Cash, 6 Holdings, 6 Prices, 8×6 Tech Indicators, 3 Additional Market Features]
Technical Indicators (8)	MACD, BOLL_UB, BOLL_LB, RSI_30, CCI_30, DX_30, SMA_30, SMA_60
State Normalization	None (Raw values are used, with internal cash scaling)
Action Space	Continuous Box space, Dimension 6
Action Meaning	Values mapped to $[-1, 1]$. Positive = Buy, Negative = Sell
Maximum Shares (h_{max})	100 shares per transaction
Trading Parameters	
Initial Account Balance	\$100,000
Transaction Fee Rate	0.01% (0.0001) for both buy and sell orders
Reward Function	Stepwise change in portfolio value plus a 10-day look-back Sharpe reward:
	$R_t = r_t^{\text{step}} + r_t^{\text{lookback}}, \quad r_t^{\text{step}} = A_t - A_{t-1}, \quad r_t^{\text{lookback}} = \frac{\Delta A_{t-10:t}}{\sqrt{\text{Var}(\Delta A_{t-10:t})} + \epsilon}$ $\Delta A_{t-10:t} = \frac{1}{10} \sum_{\tau=t-9}^t \Delta A_{\tau}, \quad \Delta A_{\tau} = A_{\tau} - A_{\tau-1} \quad \epsilon > 0 \text{ is a small constant to avoid division by zero.}$
Reward Scaling	1×10^{-4}

implementation, this look-back reward adjustment helps the agent consider short-term risk-adjusted performance in addition to immediate asset gains.

6.2.2 Quantum Circuit Architecture and Network Topology

In our FPQC-SAC implementation, the quantum component acts specifically as a *Quantum Feature Extractor* that processes the raw market state before passing the representations to the classical Actor and Critic heads.

The topology follows a bottleneck structure:

- Pre-Net (Classical):** A linear projection with ReLU activation maps the 64-dimensional state to 7 dimensions ($64 \rightarrow 7$).
- PQC Layer (Quantum):** The 7-dimensional vector is encoded into 7 qubits via Angle Embedding (R_x rotations). This is followed by 2 layers of `BasicEntanglerLayers` (parameterized rotations connected by circular CNOTs). The output is obtained via Pauli-Z expectation values ($\langle Z \rangle$) across all 7 qubits.
- Post-Net (Classical):** A linear projection with ReLU activation expands the 7-dimensional quantum measurement back to a 64-dimensional feature vector ($7 \rightarrow 64$).
- Actor-Critic Heads:** Standard SB3 MLP networks (two layers of 256 units: [256, 256]) process the 64-dimensional feature vector to output policies and Q-values

Table 7: Quantum Parameterized Quantum Circuit (PQC) settings.

Parameter	Configuration
Number of Qubits (N)	7
Encoding Strategy	<code>qml.AngleEmbedding</code>
Variational Layers	2 layers of <code>qml.BasicEntanglerLayers</code>
Observable	Expectation value of Pauli-Z ($\langle Z \rangle$) per qubit
Simulator Backend	<code>default.qubit</code> (PennyLane)
Differentiation Method	Exact backpropagation (<code>diff_method="backprop"</code>)
Total PQC Trainable Weights	14 parameters (Shape: 2×7)

6.2.3 Training Hyperparameters

The models are trained over the period from **2013-01-02 to 2018-12-31** and strictly evaluated out-of-sample from **2019-01-02 to 2021-08-31** (~ 672 trading days). Each agent is trained for 25,000 environment steps. All DRL experiments are executed across 20 distinct, fixed random seeds to ensure statistical reliability.

Table 8: Algorithmic hyperparameters for standard and quantum DRL baselines.

Hyperparameter	Value
Total Environment Steps	25,000
Learning Rate	3×10^{-4}
Batch Size (SAC, DDPG, TD3)	64
Replay Buffer Size	200,000
Discount Factor (γ)	0.99
Target Smoothing Coefficient (τ)	0.005
Entropy Coefficient (SAC/FPQC-SAC)	“auto” (dynamically adjusted to target entropy -6)
Optimizer	Adam

6.3 Extended Theoretical Analysis and Proofs

6.3.1 Proof of Proposition 1: Expectation Attenuation

Proof. Given the effective latent SDE $dZ_s = \mu_Z ds + \sigma_Z dW_s$ (from Lemma 6.3.3) over $s \in [t, t + \Delta t]$. The quantum angle encoding maps this state into complex periodic components. Let $v_\omega(Z_s) = \exp(i\omega^\top Z_s)$. Applying Itô’s formula to the complex-valued process v_ω :

$$dv_\omega(Z_s) = v_\omega(Z_s) \left(i\omega^\top \mu_Z - \frac{1}{2} \omega^\top \Sigma_Z \omega \right) ds + v_\omega(Z_s) i\omega^\top \sigma_Z dW_s \quad (7)$$

Taking the expectation $\mathbb{E}[\cdot]$ over local Brownian paths, the Itô integral with respect to Brownian motion is a martingale and evaluates to zero. Solving the deterministic ordinary differential equation over Δt yields:

$$\mathbb{E}[e^{i\omega^\top Z_{t+\Delta t}}] = e^{i\omega^\top Z_t} \exp \left(i \int_t^{t+\Delta t} \omega^\top \mu_Z ds \right) \exp \left(-\frac{1}{2} \int_t^{t+\Delta t} \omega^\top J_h \Sigma_X J_h^\top \omega ds \right) \quad (8)$$

This establishes that despite the unconstrained affine mapping J_h , the complex phase mathematically enforces an expectation-level exponential attenuation of the diffusion variance.

This section rigorously formalizes the theoretical framework established in Section 3, providing end-to-end mathematical proofs connecting the financial Itô diffusion, the entropy trap, and the quantum representation bounds.

6.3.2 Formalizing the Financial Entropy Trap

When Q-values are perturbed by environmental noise, the soft value update operator of SAC, $\alpha \log \int \exp(Q/\alpha) da$, systematically converts zero-mean noise into a positive bias. Inspired by the analysis of the log-sum-exp overestimation tendency by Kuznetsov et al. [5], we contextualize this mechanism within the bootstrapped financial low-SNR setting, identifying it as the structural root of SAC’s systematic failure in such environments.

Let the true action-value be $Q^*(s, a)$. Due to environmental diffusion, the estimated Q-value contains a zero-mean perturbation: $\tilde{Q}(s, a) = Q^*(s, a) + \epsilon(s, a)$, with $\mathbb{E}_\epsilon[\epsilon(s, a)] = 0$. The soft value estimate over the continuous action space $\mathcal{A}$ is given by $V(s) = \alpha \log \int_{\mathcal{A}} \exp(\tilde{Q}(s, a)/\alpha) da$.

Because the continuous $\log \int \exp(\cdot)$ operator is strictly convex, applying Jensen’s inequality with respect to the noise distribution yields a strict positive bias:

$$\begin{aligned}\mathbb{E}_\epsilon[V(s)] &= \mathbb{E}_\epsilon \left[\alpha \log \int_{\mathcal{A}} \exp \left(\frac{Q^*(s, a) + \epsilon(s, a)}{\alpha} \right) da \right] \\ &> \alpha \log \int_{\mathcal{A}} \exp \left(\frac{\mathbb{E}_\epsilon[Q^*(s, a) + \epsilon(s, a)]}{\alpha} \right) da \\ &= \alpha \log \int_{\mathcal{A}} \exp \left(\frac{Q^*(s, a)}{\alpha} \right) da \equiv V^*(s).\end{aligned}\tag{9}$$

Furthermore, assuming the environmental noise $\epsilon(s, a)$ is locally sub-Gaussian with a variance proxy σ_ϵ^2 , the moment-generating function restricts this overestimation bias to a deterministic upper bound :

$$0 < \mathbb{E}_\epsilon[V(s)] - V^*(s) \leq \frac{\sigma_\epsilon^2}{2\alpha}.\tag{10}$$

Mathematical Implication: These inequalities establish a harsh reality: *even if the noise is strictly zero-mean, the expected value estimation is deterministically skewed upward.* Within SAC’s bootstrapping iterations, this positive bias is repeatedly injected into the target network and continuously amplified, forming the mathematical genesis of the Financial Entropy Trap. The upper bound $\mathcal{O}(\sigma_\epsilon^2/\alpha)$ explicitly dictates that the larger the market diffusion variance σ_ϵ^2 , the more severe the overestimation; furthermore, as the entropy temperature α decays during training, this explosive bias is further exacerbated.

Financial Intuition: The consequence of this mathematical mechanism can be intuitively understood through a notorious market phenomenon. In a highly uncertain environment, asset price fluctuations inherently lack clear directional signals. However, some traders may misinterpret a random upward noise spike as a genuine trend signal and choose to follow. These follow-up actions artificially drive the price higher, attracting even more participants and forming a self-reinforcing positive feedback loop.

Yet, when the price inevitably reverts to its fundamentals, the accumulated directional positions reverse almost synchronously. The resulting stampede causes a drawdown far exceeding the initial artificial gains. Late participants not only fail to realize their paper profits but suffer severe real losses. This market dynamic perfectly mirrors the catastrophic policy degradation of an RL agent falling into the Financial Entropy Trap.

6.3.3 Pre-Net Jacobian and Local Effective Itô Diffusion

We use a standard local reduction to connect the raw Itô state process to the latent diffusion analyzed in Appendix 6.3.1. Prior work in continuous-time finance has established the use of neural-network Jacobians and automatic differentiation together with Itô’s lemma for high-dimensional stochastic dynamic systems [14]. Separately, SDE-learning literature has observed that when the transformation network is piecewise linear, such as ReLU networks, the second-order Itô correction vanishes almost everywhere, reducing the transformed dynamics to Jacobian-vector products [10].

Concretely, let the raw state follow

$$dX_t = \mu_X dt + \sigma_X dW_t,\tag{11}$$

and let the preprocessed latent state be

$$Z_t = h_{\text{pre}}(X_t).\tag{12}$$

Since the Pre-Net uses piecewise-linear activations such as ReLU, it is locally affine within each fixed activation region. Away from nonsmooth activation boundaries, the diffusion component is therefore locally propagated through the Pre-Net Jacobian:

$$dZ_t \approx J_h(X_t) \mu_X dt + J_h(X_t) \sigma_X dW_t,\tag{13}$$

where $J_h(X_t)$ denotes the local Jacobian of h_{pre} at X_t . Consequently, the local instantaneous covariance of the latent process is

$$\Sigma_Z(X_t) = (J_h(X_t) \sigma_X) (J_h(X_t) \sigma_X)^\top = J_h(X_t) \Sigma_X J_h(X_t)^\top.\tag{14}$$

This step is a standard local piecewise-linear approximation used to identify the effective diffusion covariance entering the PQC encoder. A fully exact treatment of nonsmooth activation boundaries would require generalized Itô–Tanaka arguments and may introduce local-time correction terms at kink boundaries. Since our subsequent analysis only requires the local covariance propagated into the latent space, we use $\Sigma_Z = J_h \Sigma_X J_h^\top$ as the effective latent diffusion covariance.

6.3.4 The Role of Entanglement: Cross-Dimensional Feature Crossing and Constructive Interference of Consensus Signals

A Parameterized Quantum Circuit (PQC) without CNOT gates restricts Pauli-Z measurements strictly to separable single-coordinate Fourier modes. In financial markets, this is equivalent to relying on a single technical indicator, which is often noisy and prone to failure. In contrast, CNOT entanglement forces local single-qubit states to bind into multi-body Pauli strings, thereby generating inseparable cross-coordinate feature interactions. This entangled structure acts as an efficient high-dimensional feature crossing mechanism, enabling the model to evaluate joint signals across assets or indicators: only when multiple weak signals form a consistent structural consensus within the entangled state are they amplified through measurement (constructive interference); meanwhile, isolated, uncorrelated random noise cancels out in the complex phase superposition (destructive interference).

To illustrate this mechanism structurally, consider that in a circuit without entanglement, the unitary evolution factorizes completely across qubits:

$$U(Z) = \bigotimes_{j=1}^n U_j(Z_j).$$

For a local Pauli-Z measurement on qubit j , the expectation takes the form

$$f_j(Z) = \langle 0 | U_j(Z_j)^\dagger Z U_j(Z_j) | 0 \rangle,$$

and therefore depends only on the single latent coordinate Z_j . Under angle encoding, its Fourier spectrum is confined to single-coordinate modes such as $\omega = \pm e_j$. From a quantitative trading perspective, this corresponds to the model making decisions based solely on a single isolated indicator (such as an individual MACD or RSI), which frequently fails in low-SNR markets. By Proposition 1, these modes undergo expectation attenuation governed by:

$$\exp \left(-\frac{1}{2} \int_t^{t+\Delta t} e_j^\top \Sigma_Z e_j ds \right).$$

This implies that without entanglement, phase attenuation occurs strictly independently along individual coordinate axes, and traditional linear feature extraction struggles to capture high-order correlations.

Now consider a CNOT gate with control qubit j and target qubit k . In the Heisenberg picture:

$$\text{CNOT}_{j \rightarrow k}^\dagger (I \otimes Z_k) \text{CNOT}_{j \rightarrow k} = Z_j \otimes Z_k.$$

Thus, a local measurement after the entangling layer can correspond to a two-body Pauli string before the entangler. Its expectation contains products of trigonometric phase terms, for example:

$$\cos(Z_j) \cos(Z_k) = \frac{1}{2} [\cos(Z_j + Z_k) + \cos(Z_j - Z_k)].$$

Through this mechanism, entanglement naturally introduces joint Fourier modes:

$$\omega = e_j + e_k, \quad \omega = e_j - e_k.$$

These modes also inherit the attenuation law in Proposition 1:

$$\exp \left(-\frac{1}{2} \int_t^{t+\Delta t} (e_j \pm e_k)^\top \Sigma_Z (e_j \pm e_k) ds \right).$$

Expanding the quadratic form gives:

$$(e_j \pm e_k)^\top \Sigma_Z (e_j \pm e_k) = (\Sigma_Z)_{jj} + (\Sigma_Z)_{kk} \pm 2(\Sigma_Z)_{jk}.$$

681 This demonstrates that entanglement does not merely blindly add more non-linear features. It
682 fundamentally provides an efficient cross-dimensional feature crossing mechanism. When processing
683 financial features, this mechanism exhibits distinctly different filtering properties for "noise" versus
684 "signal":

685 **1. Destructive Interference of Isolated Noise (Analogous to Denoising):** For independent, isotropic
686 noise lacking intrinsic correlation (where $\Sigma_Z = \sigma^2 I$), the joint mode satisfies:

$$(e_j \pm e_k)^\top \Sigma_Z (e_j \pm e_k) = 2\sigma^2,$$

687 which is larger than the single-coordinate variance σ^2 . This indicates that unstructured random noise
688 lacking correlation will induce more violent phase fluctuations within these entangled cross modes,
689 thereby accelerating attenuation and canceling each other out in the final expectation projection.

690 **2. Constructive Interference of Structural Consensus (Analogous to Signal Amplification):**
691 Conversely, effective trading signals are often hidden within the joint distributions or non-linear
692 combinations of multiple indicators (e.g., price breakouts accompanied by specific volume and
693 volatility contractions). When there is a weak drift structure shared across adjacent latent coordinates
694 (i.e., a genuine market consensus exists among indicators), it induces coherent phase translations in
695 the related qubits. Trainable rotation gates can dynamically adjust the measurement basis so that
696 these structural consensus signals (correlated phase translations) are precisely projected onto selected
697 Pauli readout directions. Along these directions, the phase contributions of multiple weak signals
698 undergo constructive accumulation (constructive interference) and are effectively amplified.

699 In conclusion, entanglement doesn't help by magically cleaning up noise in a perfect mathematical
700 sense. Instead, it forces independent indicators to coalesce into multi-body states. This setup not only
701 uses the diffusion attenuation mechanism from Proposition 1 to keep uncorrelated noise in check,
702 but also adds the ability to capture—and amplify—the structural consensus embedded in financial
703 signals.

704 6.4 Weak-Signal-Preserving Property

705 **Lemma 1** (Representation Output SNR Lower Bound). *Consider an environment transition $X \rightarrow$
706 $X + \delta + \xi$, where δ is a weak deterministic drift signal and ξ is high-variance zero-mean noise. Let*

$$J_\phi^{\text{eff}}(X) = \nabla_X \mathbb{E}_\xi[\phi(X + \xi)]$$

707 *denote the effective local Jacobian of the expectation-level PQC mapping. If the trained circuit is*
708 *non-singular along the signal direction such that*

$$\sigma_{\min}(J_\phi^{\text{eff}}|_{\text{span}(\delta)}) \geq c > 0,$$

709 *then for any fixed nonzero drift δ , the output signal-to-noise ratio is lower-bounded by*

$$\text{SNR}_{\text{out}} \geq \frac{c^2 \|\delta\|_2^2}{k}.$$

710 *Proof.* By a first-order Taylor expansion of the expected representation mapping, the preserved signal
711 magnitude satisfies

$$\|\Delta \mathbb{E}[\phi]\|_2^2 = \|\mathbb{E}[\phi(X + \delta + \xi)] - \mathbb{E}[\phi(X + \xi)]\|_2^2 \approx \|J_\phi^{\text{eff}}(X)\delta\|_2^2 \geq c^2 \|\delta\|_2^2.$$

712 The output noise power is measured by the trace of the covariance across the representation dimen-
713 sions:

$$\text{Tr}(\text{Cov}[\phi(X + \xi)]) = \sum_{j=1}^k \text{Var}(\phi_j(X + \xi)).$$

714 By the bounded Pauli readout established in Proposition 2, each PQC feature dimension satisfies
715 $\phi_j(X + \xi) \in [-1, 1]$. Therefore, by Popoviciu's inequality,

$$\text{Var}(\phi_j(X + \xi)) \leq \frac{(1 - (-1))^2}{4} = 1.$$

Consequently,

$$\text{Tr}(\text{Cov}[\phi(X + \xi)]) \leq k.$$

The output SNR therefore satisfies

$$\text{SNR}_{\text{out}} = \frac{\|\Delta\mathbb{E}[\phi]\|_2^2}{\text{Tr}(\text{Cov}[\phi])} \geq \frac{c^2\|\delta\|_2^2}{k}.$$

This shows that the PQC bottleneck can preserve a nonzero deterministic signal component along learned non-singular directions, while imposing a dimension-dependent ceiling on the output noise power.

6.5 Extended Out-of-Sample Extrapolation (2019–2024)

To further evaluate the long-term robustness and generalization capability of the learned policies, we conduct an extended out-of-sample extrapolation experiment. In this setting, the DRL agents, which were trained solely on data up to 2018-12-31, are deployed without any retraining over an extended testing horizon from 2019-01-02 to 2023-12-31. This prolonged period introduces entirely unseen macroeconomic regimes, most notably the sustained market downturn of 2022 and the subsequent recovery phase in 2023.

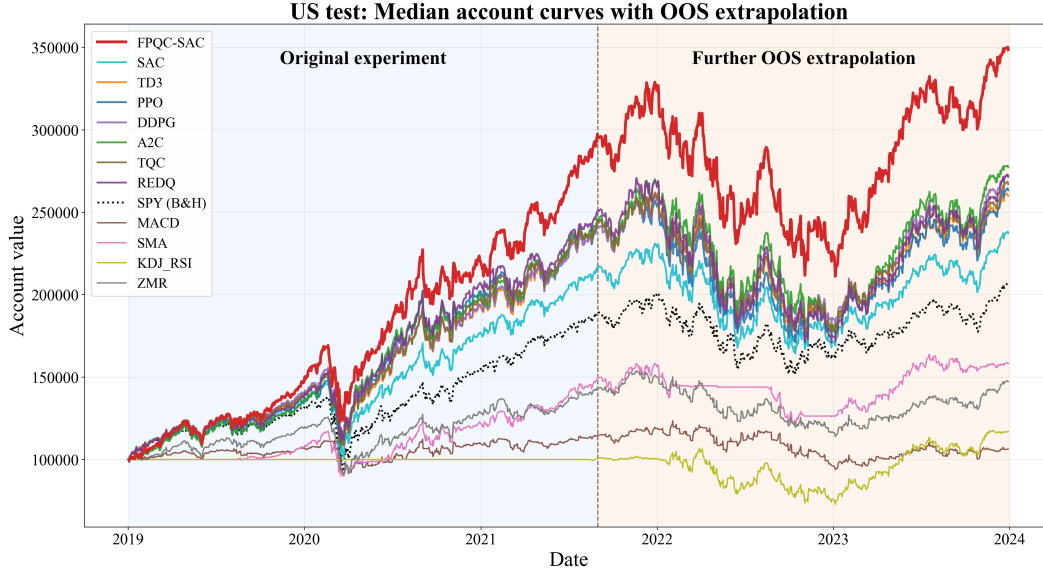


Figure 4: Extended out-of-sample median account value trajectories from 2019 to 2023. The shaded area represents the further extrapolation period beyond the original experiment. FPQC-SAC (red) maintains its relative rank and overall trajectory without structural breakdown despite the lack of retraining over the extended horizon.

727

Figure 4 illustrates the median account value trajectories across this extended horizon. As expected, the systemic macroeconomic shifts in 2022 induce drawdowns across all evaluated models. However, FPQC-SAC consistently maintains its relative positioning above the DRL baselines. The policy does not exhibit catastrophic forgetting or structural breakdown over the prolonged out-of-sample period, remaining resilient even when the market environment deviates from the 2013–2018 training distribution.

6.6 Statistical Significance Analysis

To rigorously evaluate whether the observed performance improvements are statistically reliable, we conduct Welch’s t-tests comparing FPQC-SAC against SAC across 20 independent random seeds. The results are detailed in the following table:

Table 9: Welch’s t-test significance analysis comparing FPQC-SAC against SAC

Dataset	Metric	Welch’s p-value	Significance Level	Statistical Confidence
Mainstream Tech & Market-Index	Cumulative Return	4.14×10^{-10}	***	>99.99%
	Annual Return	1.11×10^{-10}	***	>99.99%
	Sharpe Ratio	3.74×10^{-7}	***	>99.99%
	Sortino Ratio	3.87×10^{-9}	***	>99.99%
	Calmar Ratio	2.96×10^{-9}	***	>99.99%
Defensive Blue-chip (Sub-experiment 1)	Cumulative Return	1.87×10^{-2}	*	>98.13%
	Annual Return	2.06×10^{-2}	*	>97.94%
	Sharpe Ratio	4.90×10^{-2}	*	>95.10%
	Sortino Ratio	2.83×10^{-2}	*	>97.17%
	Calmar Ratio	8.72×10^{-3}	**	>99.13%

Strong statistical significance in volatile regime (Main Experiment). On the Mainstream Tech & Market-Index Portfolio (QQQ, SPY, MSFT, GOOGL, AMZN, AAPL), which represents a technology-heavy and high-volatility regime, FPQC-SAC achieves highly statistically significant improvements across all evaluation metrics. All p-values are well below 10^{-6} , corresponding to confidence levels exceeding 99.99% (***). This provides strong evidence that the observed performance gains are remarkably robust and not attributable to random variation, validating the effectiveness of FPQC-SAC in low-SNR environments with strong stochastic perturbations.

Consistent significance in stable regime (Sub-experiment 1). In the Defensive Blue-chip Portfolio (Sub-experiment 1), FPQC-SAC also successfully reaches statistical significance across all metrics. The Calmar Ratio shows strong significance ($p < 0.01$), while the Cumulative Return, Annual Return, Sharpe Ratio, and Sortino Ratio all achieve conventional significance thresholds ($p < 0.05$). This is a noteworthy achievement, as stable blue-chip environments typically compress the performance margin between competing methods.

Interpretation: Robustness across diverse market conditions. In the stable blue-chip setting, many Deep Reinforcement Learning (DRL) baselines struggle to consistently outperform the Buy-and-Hold (B&H) benchmark, as the optimal policy is close to a near-static allocation. Consequently, the achievable performance margin is inherently limited, and effect sizes are often small relative to residual variance. Achieving statistical significance in this regime demonstrates that FPQC-SAC’s improvements are not only directionally consistent but also systematically reliable.

Implication. These findings highlight a key property of FPQC-SAC: it provides universally robust improvements over SAC. Its advantage is highly pronounced and undeniable in challenging environments where dynamic decision-making under noise is essential. More importantly, even in stable, low-variance markets—where static strategies already perform competitively and algorithmic edges are notoriously difficult to prove—FPQC-SAC still delivers statistically significant gains. Overall, the presented significance analysis confirms FPQC-SAC as a highly effective and reliable framework across varying market dynamics.

6.7 Evaluation Metrics

To comprehensively assess the trading performance and risk management capabilities of the learned policies, we employ five standard financial evaluation metrics.

Before defining the metrics, we first formally define the portfolio net return. Let V_t denote the total portfolio value at time step t . At each step, the agent executes trading actions that incur a transaction fee rate of $c = 0.01\%$ (0.0001) for both buying and selling. The net daily return R_t at step t naturally incorporates this transaction cost and is given by:

$$R_t = \frac{V_t - V_{t-1}}{V_{t-1}} \quad (15)$$

where V_t is the net portfolio value after deducting the transaction fees associated with the generated action a_t . Let T be the total number of trading days in the out-of-sample period, and R_f be the daily risk-free return rate (set to zero in our experiments for simplicity). The metrics are defined as follows:

1. Cumulative Return (CR%) Cumulative Return measures the overall percentage increase of the portfolio value from the beginning to the end of the trading period.

$$CR = \frac{V_T - V_0}{V_0} \times 100\% \quad (16)$$

2. Annualized Return (AR%) Annualized Return represents the geometric average amount of money earned by the investment each year over a given time period. Assuming 252 trading days in a year, it is calculated as:

$$AR = \left((1 + CR)^{\frac{252}{T}} - 1 \right) \times 100\% \quad (17)$$

3. Sharpe Ratio (SR) The Sharpe Ratio evaluates the risk-adjusted return of the portfolio by penalizing excessive volatility. It is the ratio of the annualized expected excess return to the annualized standard deviation of returns:

$$SR = \frac{\sqrt{252} \cdot \mathbb{E}[R_t - R_f]}{\sigma(R_t)} \quad (18)$$

where $\mathbb{E}[R_t - R_f]$ is the mean of the daily excess returns, and $\sigma(R_t)$ is the standard deviation of the daily returns.

4. Sortino Ratio (Sortino) The Sortino Ratio is a variation of the Sharpe ratio that differentiates harmful volatility from total overall volatility by using the asset's downside deviation. It only penalizes negative returns:

$$\text{Sortino} = \frac{\sqrt{252} \cdot \mathbb{E}[R_t - R_f]}{\sigma_d} \quad (19)$$

where $\sigma_d = \sqrt{\mathbb{E}[\min(0, R_t - R_f)^2]}$ is the target downside deviation.

5. Calmar Ratio (Calmar) The Calmar Ratio measures the annualized return relative to the Maximum Drawdown (MDD), serving as an indicator of return relative to tail-end downside risk. The MDD represents the largest peak-to-trough drop in the portfolio value:

$$MDD = \max_{\tau \in (0, T)} \left(\frac{\max_{t \in (0, \tau)} V_t - V_\tau}{\max_{t \in (0, \tau)} V_t} \right) \quad (20)$$

$$\text{Calmar} = \frac{AR}{MDD} \quad (21)$$

6.8 Limitation And Future Work

While FPQC-SAC demonstrates substantial performance leaps in low-SNR financial environments, we outline the theoretical and computational boundaries of the current framework, which naturally motivate our future research trajectory.

Theoretical Boundaries and Degenerate Market Regimes. The noise-attenuation mechanisms of FPQC-SAC are fundamentally anchored in the assumption that market dynamics exhibit Itô-diffusion characteristics (as defined in Assumption 1). Specifically, it excels at separating high-variance Brownian perturbations from weak but existent deterministic drift (predictive signals). Consequently, the efficacy of this denoising paradigm may be intrinsically bounded in fully degenerate market regimes—such as assets driven purely by macroscopic behavioral sentiment, meme-stock frenzies, or zero-drift micro-structure noise. In such scenarios, the lack of an underlying structural signal renders any signal-preserving operation mathematically less meaningful, as the market itself behaves as pure noise. We view this not as an algorithmic deficiency, but rather as a rigorous reflection of the model's theoretical specificity: FPQC-SAC is designed to extract weak signals where they exist, rather than hallucinate signals where there are none.

Computational Footprint and Scalability. In this study, our empirical validation focuses on highly representative, compact asset portfolios. This design choice is partially motivated by the inherent physical constraints of classical hardware: simulating the entanglement and unitary evolution of high-dimensional quantum states entails an exponentially growing computational footprint. Consequently, extending the current classical-simulation-based FPQC-SAC to massively scaled combinatorial portfolios (e.g., thousands of concurrent assets) presents practical computational bottlenecks. However, our strategic use of the PQC as a *compact bottleneck* elegantly turns this limitation into an advantage, enabling efficient deployment on current classical GPUs without compromising the architectural integrity of the quantum representation.

Future Work: Transitioning to Physical Quantum Hardware. The computational bottleneck of simulating quantum mechanics on classical machines is precisely what quantum hardware is designed to resolve. As an exciting and natural progression, our immediate future work aims to deploy the FPQC-SAC architecture onto real-world physical Quantum Processing Units (QPUs). By transitioning from classical simulators to actual quantum hardware, we anticipate bypassing the exponential simulation overhead. This will allow us to scale the qubit register to accommodate massive-scale portfolio management tasks, fully unlocking the intrinsic parallel processing power and natural entanglement dynamics of genuine quantum systems in real-time financial trading.

6.9 Border Impact

This paper presents work whose goal is to advance the field of Finance and Artificial Intelligence. We will open-source the code and accept public review in the future. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

NeurIPS Paper Checklist

1. Claims

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Answer: [\[Yes\]](#)

Justification: The abstract and introduction (Section 1) identify the "Financial Entropy Trap" and propose the FPQC-SAC architecture. These claims are supported by theoretical derivations in Section 2, Appendix 5.3, and the extensive empirical evaluations in Section 3 and Table 1-3.

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2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

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Justification: Section 5.8 outlines limitations regarding the "Itô-diffusion" assumption in degenerate market regimes and the current computational scalability bottleneck when simulating high-dimensional quantum states on classical hardware.

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Justification: All assumptions are formalized in Section 2.1 (Assumption 1). Detailed proofs for the entropy trap bias (Eq. 11-12), geometric constraints (Proposition 1), and SNR lower bounds (Lemma 1) are provided in the Appendix (Section 5.3 and 5.4).

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Justification: Section 5.2 provides a detailed MDP formulation (Table 6), the quantum circuit topology (Table 7), and algorithmic hyperparameters (Table 8) used for all DRL baselines and the proposed method.

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Justification: Detailed training/test periods (2013-2021) are provided in Section 3.1 and 5.2.3. Hyperparameters for SAC, DDPG, and TD3 are listed in Table 8. The choice of Adam optimizer is also noted.

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